Report OOP – Lab 04

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I. New written code

1. Tạo lớp Book

Branch: topic/aims-project/add-book-class

Tạo Class **Book** trong package **hust.soict.hedspi.aims.media** .

```java
package hust.soict.hedspi.aims.media;

import java.util.*;

public class Book {

    private int id;
    private String title;
    private String category;
    private float cost;
    private List<String> authors = new ArrayList<String>();


    public int getId() {
        return id;
    }


    public void setId(int id) {
        this.id = id;
    }


    public String getTitle() {
        return title;
    }


    public void setTitle(String title) {
        this.title = title;
    }


    public String getCategory() {
        return category;
    }


    public void setCategory(String category) {
        this.category = category;
    }
```

```java
        this.category = category;
    }


    public float getCost() {
        return cost;
    }


    public void setCost(float cost) {
        this.cost = cost;
    }


    public List<String> getAuthors() {
        return authors;
    }


    public void setAuthors(List<String> authors) {
        this.authors = authors;
    }


    public Book(String title) {
        this.title = title;
    }

    public Book(String title, String category) {
        this.title = title;
        this.category = category;
    }

    public Book(String title, String category, float cost) {
        this.title = title;
        this.category = category;
        this.cost = cost;
    }

    public void addAuthor(String authorName) {
        if(!authors.contains(authorName)) {
```

Tạo các phương thức **addAuthor(String authorName)** và **removeAuthor(String authorName)** cho lớp **Book**:

```java
    public void addAuthor(String authorName) {
        if(!authors.contains(authorName)) {
            authors.add(authorName);
            System.out.println("Add author successfully!");
        } else {
            System.out.println("This author has already been in the list!");
        }
    }

    public void removeAuthor(String authorName) {
        if (authors.contains(authorName)) {
            authors.remove(authorName);
            System.out.println("Remove author successfully!");
        } else {
            System.out.println("No author has been found to remove!");
        }
    }

}
```

2. Tạo lớp trừu tượng Media

Branch: topic/aims-project/add-media-class

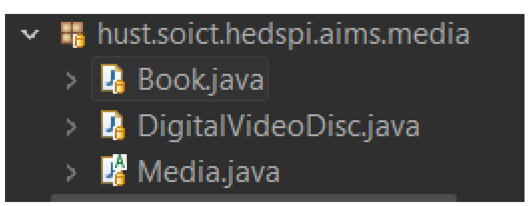
Tạo lớp trừu tượng **Media** có chứa các thông tin chung của **DVD** và **Book**.

```java
package hust.soict.hedspi.aims.media;

public abstract class Media {

    private static int nbMedia = 0;
    private int id;
    private String title;
    private String category;
    private float cost;


    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getTitle() {
        return title;
    }

    public void setTitle(String title) {
        this.title = title;
    }

    public String getCategory() {
        return category;
    }

    public void setCategory(String category) {
        this.category = category;
    }

    public float getCost() {
        return cost;
    }

    public void setCost(float cost) {
        this.cost = cost;
```

```java

    public void setTitle(String title) {
        this.title = title;
    }

    public String getCategory() {
        return category;
    }

    public void setCategory(String category) {
        this.category = category;
    }

    public float getCost() {
        return cost;
    }

    public void setCost(float cost) {
        this.cost = cost;
    }

    public Media(String title) {
        this.title = title;
        this.id = ++nbMedia;
    }

    public Media(String title, String category) {
        this.title = title;
        this.category = category;
        this.id = ++nbMedia;
    }

    public Media(String title, String category, float cost) {
        this.title = title;
        this.category = category;
        this.cost = cost;
        this.id = ++nbMedia;
    }

}

```

Đồng thời, xóa các trường và phương thức khỏi lớp **Book** và **DigitalVideoDisc**.

Chuyển **DigitalVideoDisc** vào package **hust.soict.hedspi.aims.media** và xóa package **hust.soict.dsai.aims.disc**



Kế thừa lớp **Media** cho cả 2 lớp **Book** và **DigitalVideoDisc**.

- Lớp **Book**:

Media.java Book.java

```java
public class Book extends Media {

    private List<String> authors = new ArrayList<String>();

    public List<String> getAuthors() {
        return authors;
    }

    public void setAuthors(List<String> authors) {
        this.authors = authors;
    }

    public Book(String title) {
        super(title);
    }

    public Book(String title, String category) {
        super(title, category);
    }

    public Book(String title, String category, float cost) {
        super(title, category, cost);
    }

    public void addAuthor(String authorName) {
        if(!authors.contains(authorName)) {
            authors.add(authorName);
            System.out.println("Add author successfully!");
        } else {
            System.out.println("This author has already been in the list!");
        }
    }

    public void removeAuthor(String authorName) {
        if (authors.contains(authorName)) {
            authors.remove(authorName);
            System.out.println("Remove author successfully!");
        } else {
            System.out.println("No author has been found to remove!");
        }
    }
```

- Lớp **DigitalVideoDisc**

```java
public class DigitalVideoDisc extends Media {

    private String director;
    private int length;

    public String getDirector() {
        return director;
    }
    public int getLength() {
        return length;
    }

    public boolean isMatch(String keyword)
    {
        return this.getTitle().toLowerCase().contains(keyword.toLowerCase());
    }

    public DigitalVideoDisc(String title) {
        super(title);
    }
    public DigitalVideoDisc(String title, String category, float cost) {
        super(title, category, cost);
    }
    public DigitalVideoDisc(String title, String category, String director, float cost) {
        super(title, category, cost);
        this.director = director;
    }
    public DigitalVideoDisc(String title, String category, String director, int length, float cost) {
        super(title, category, cost);
        this.director = director;
        this.length = length;
    }

    @Override
    public String toString()
    {
        return "DVD: " + this.getTitle() +
                " - Category: " + this.getCategory() +
                " - Director: " + this.director +
```

3. Tạo lớp CompactDisc

Branch: topic/aims-project/create-CompactDisc-class

3.1. Tạo lớp Disc kế thừa từ Media

Tạo lớp **Disc** có 2 trường length và director.

Làm cho **DigitalVideoDisc** và **CompactDisc** kế thừa từ lớp **Disc**.

- Lớp **Disc**

Media.java Book.java DigitalVideoDisc.java Disc.java

```java
package hust.soict.hedspi.aims.media;

public class Disc extends Media {

    private String director;
    private int length;

    public String getDirector() {
        return director;
    }

    public void setDirector(String director) {
        this.director = director;
    }

    public int getLength() {
        return length;
    }

    public void setLength(int length) {
        this.length = length;
    }

     public Disc(String title) {
        super(title);
    }

    public Disc(String title, String category) {
        super(title, category);
    }

    public Disc(String title, String category, float cost) {
        super(title, category, cost);
    }

    public Disc(String title, String category, String director, float cost) {
        super(title, category, cost);
        this.director = director;
    }

    public Disc(String title, String category, String director, int length, float cost) {
```

- Lớp **DigitalVideoDisc**

Media.java | Book.java | DigitalVideoDisc.java | Disc.java

```java
package hust.soict.hedspi.aims.media;

public class DigitalVideoDisc extends Disc {

    public boolean isMatch(String keyword)
    {
        return this.getTitle().toLowerCase().contains(keyword.toLowerCase());
    }

    public DigitalVideoDisc(String title) {
        super(title);
    }
    public DigitalVideoDisc(String title, String category, float cost) {
        super(title, category, cost);
    }
    public DigitalVideoDisc(String title, String category, String director, float cost) {
        super(title, category, director, cost);
    }
    public DigitalVideoDisc(String title, String category, String director, int length, float cost) {
        super(title, category, director, length, cost);
    }

    @Override
    public String toString()
    {
        return "DVD: " + this.getTitle() +
                " - Category: " + this.getCategory() +
                " - Director: " + this.getDirector() +
                " - DVD length: " + this.getLength() +
                " - Cost: " + this.getCost() + "$";
    }

}
```

3.2. Tạo lớp Track

Lớp **Track** mô phỏng một bài nhạc trên đĩa **CompactDisc**.

Media.java | Book.java | DigitalVideoDisc.java | Disc.java | Track.java

```java
package hust.soict.hedspi.aims.media;

public class Track {

    private String title;
    private int length;

    public String getTitle() {
        return title;
    }

    public int getLength() {
        return length;
    }

    public Track(String title, int length) {
        this.title = title;
        this.length = length;
    }

}
```

3.3. Sửa lớp CompactDisc

Thêm trường artist và danh sách tracks.

Tạo phương thức **addTrack()** và **removeTrack().**

Tạo phương thức **getLength()** để tính tổng độ dài các bài nhạc trong tracks của CD.

Media.java | Book.java | DigitalVideoDisc.java | Disc.java | Track.java | CompactDisc.java

```java
package hust.soict.hedspi.aims.media;

import java.util.ArrayList;

public class CompactDisc extends Disc {

    private String artist;
    private ArrayList<Track> tracks;

    public String getArtist() {
        return artist;
    }

    public CompactDisc(String title) {
        super(title);
    }

    public CompactDisc(String title, String category) {
        super(title, category);
    }

    public CompactDisc(String title, String category, float cost) {
        super(title, category, cost);
    }

    public CompactDisc(String title, String category, String artist, float cost) {
        super(title, category, cost);
        this.artist = artist;
    }

    public void addTrack(Track track) {
        if(!tracks.contains(track)) {
            tracks.add(track);
            System.out.println("Add track successfully!");
        } else {
            System.out.println("Track already exists in CD.");
        }
    }

    public void removeTrack(Track track) {
        if (tracks.contains(track)) {
            tracks.remove(track);
            System.out.println("Remove track successfully!");
        } else {
            System.out.println("Track does not exist in CD.");
        }
    }

    public int getLength() {
        int totalLength = 0;
        for (Track track : tracks) {
            totalLength += track.getLength();
        }
        return totalLength;
    }

}
```

4. Tạo Interface Playable

Branch: topic/aims-project/playable-interface

Interface **Playable** cho phép các lớp chỉ định chúng triển khai phương thức **play()**

- Tạo interface **Playable** và thêm phương thức nguyên mẫu:

```java
package hust.soict.hedspi.aims.media;

public interface Playable {

    public void play();

}
```

- Triển khai interface **Playable** và phương thức **play()** trong **DigitalVideoDisc**

```java
public class DigitalVideoDisc extends Disc implements Playable {
```

```java
    public void play() {
        System.out.println("Playing DVD: " + this.getTitle());
        System.out.println("DVD length: " + this.getLength());
    }
```

- Triển khai interface **Playable** và phương thức **play()** trong **Track**

```java
public class Track implements Playable {
```

```java
    public void play() {
        System.out.println("Playing track: " + this.getTitle());
        System.out.println("Track length: " + this.getLength());
    }
```

- Triển khai interface **Playable** và phương thức **play()** trong **CompactDisc**

```java
public class CompactDisc extends Disc implements Playable {
```

```java
    public void play() {
        System.out.println("Playing CD: " + this.getTitle());
        System.out.println("CD length: " + this.getLength());
        for (Track track : tracks) {
            track.play();
        }
    }
```

5. Cập nhật lớp Cart để làm việc với Media

Branch: feature/aims-project/update-cart-to-work-with-media

Xóa các phương thức **addDigitalVideoDisc()** và **removeDigitalVideoDisc()** để tạo các phương thức tổng quát hơn là **addMedia()** và **removeMedia().**

```java
package hust.soict.hedspi.aims.cart;

import java.util.*;
import hust.soict.hedspi.aims.media.Media;

public class Cart {

    public static final int MAX_NUMBERS_ORDERED = 20;
    private ArrayList<Media> itemsOrdered = new ArrayList<Media>();

    public int qtyOrdered = 0;

    public void addMedia(Media media) {
        if (itemsOrdered.size() >= MAX_NUMBERS_ORDERED) {
            System.out.println("The cart is almost full!");
        } else {
            itemsOrdered.add(media);
            System.out.println(media.getTitle() + "has been added!");
        }
    }

    public void removeMedia(Media media) {
        if (itemsOrdered.size() == 0) {
            System.out.println("Nothing to remove!");
        } else {
            if (itemsOrdered.remove(media)) {
                System.out.println(media.getTitle() + "has been remove from the cart.");
            } else {
                System.out.println("Media not found in cart!");
            }
        }
    }
```

Cập nhật **totalCost()** để tính tổng chi phí tất cả Media

```java
    public float totalCost() {
        float totalCost = 0;
        for (Media media : itemsOrdered) {
            totalCost += media.getCost();
        }
        return totalCost;
    }
```

6. Cập nhật lớp Store để hoạt động với Media

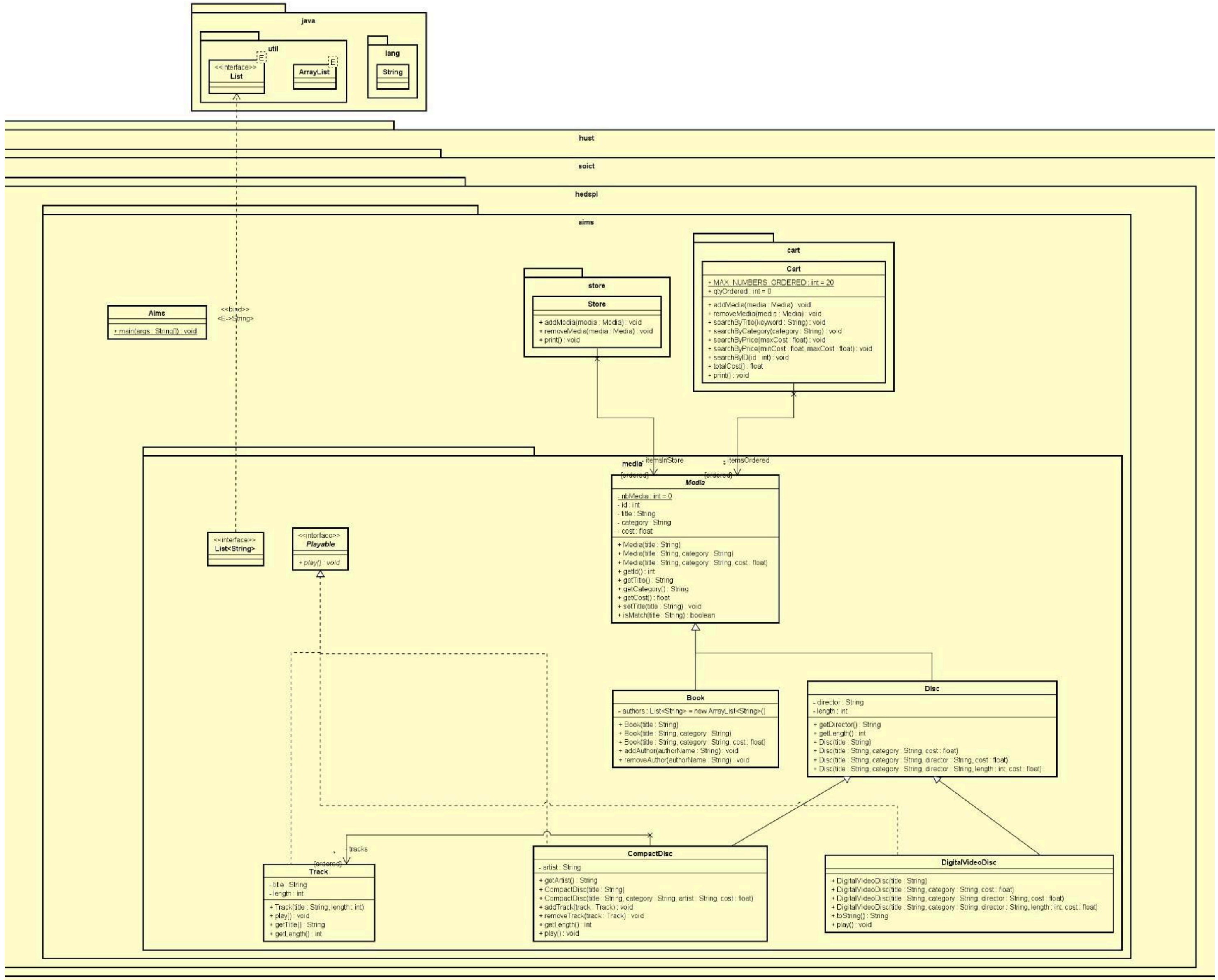
Branch: feature/aims-project/update-store-to-work-with-media

Store.java

```java
package hust.soict.hedspi.aims.store;
import java.util.ArrayList;

public class Store {

    private ArrayList<Media> itemsInStore = new ArrayList<Media>();

    public void addMedia(Media media) {
        if (itemsInStore.contains(media)) {
            System.out.println("The media " + media.getTitle() + " is already in the store!");
        } else {
            itemsInStore.add(media);
            System.out.println("The media " + media.getTitle() + " has been added to the store.");
        }
    }

    public void removeMedia(Media media) {
        if (itemsInStore.remove(media)) {
            System.out.println("The media " + media.getTitle() + " has been removed from the store.");
        } else {
            System.out.println("The media " + media.getTitle() + " is not in the store!");
        }
    }

```

7. Cập nhật sơ đồ UML và Constructor cho các lớp cha và lớp con

- Đã constructor ở các bước trước.
- Chỉ xóa những setter không cần thiết.
- Class Diagram UML



java
util
<<interface>>
List
ArrayList
lang
String
hust
soict
hedspi
aims
Aims
+ main(args : String[]) : void
<<bind>>
<E->String>
store
Store
+ addMedia(media : Media) : void
+ removeMedia(media : Media) : void
+ print() : void
cart
Cart
+ MAX_NUMBERS_ORDERED : int = 20
+ qtyOrdered : int = 0
+ addMedia(media : Media) : void
+ removeMedia(media : Media) : void
+ searchByTitle(keyword : String) : void
+ searchByCategory(category : String) : void
+ searchByPrice(maxCost : float) : void
+ searchByPrice(minCost : float, maxCost : float) : void
+ searchByID(id : int) : void
+ totalCost() : float
+ print() : void
media
itemsInStore
itemsOrdered
{ordered}
Media
- nbMedia : int = 0
- id : int
- title : String
- category : String
- cost : float
+ Media(title : String)
+ Media(title : String, category : String)
+ Media(title : String, category : String, cost : float)
+ getId() : int
+ getTitle() : String
+ getCategory() : String
+ getCost() : float
+ setTitle(title : String) : void
+ isMatch(title : String) : boolean
<<interface>>
List<String>
<<interface>>
Playable
+ play() : void
Book
- authors : List<String> = new ArrayList<String>()
+ Book(title : String)
+ Book(title : String, category : String)
+ Book(title : String, category : String, cost : float)
+ addAuthor(authorName : String) : void
+ removeAuthor(authorName : String) : void
Disc
- director : String
- length : int
+ getDirector() : String
+ getLength() : int
+ Disc(title : String)
+ Disc(title : String, category : String, cost : float)
+ Disc(title : String, category : String, director : String, cost : float)
+ Disc(title : String, category : String, director : String, length : int, cost : float)
tracks
Track
- title : String
- length : int
+ Track(title : String, length : int)
+ play() : void
+ getTitle() : String
+ getLength() : int
CompactDisc
- artist : String
+ getArtist() : String
+ CompactDisc(title : String)
+ CompactDisc(title : String, category : String, artist : String, cost : float)
+ addTrack(track : Track) : void
+ removeTrack(track : Track) : void
+ getLength() : int
+ play() : void
DigitalVideoDisc
+ DigitalVideoDisc(title : String)
+ DigitalVideoDisc(title : String, category : String, cost : float)
+ DigitalVideoDisc(title : String, category : String, director : String, cost : float)
+ DigitalVideoDisc(title : String, category : String, director : String, length : int, cost : float)
+ toString() : String
+ play() : void

8. Đảm bảo phần tử duy nhất trong danh sách Media hoặc Track

Branch: feature/aims-project/override-equals-method

Ghi đè phương thức **equals(Object o)**

- Lớp **Media**

```
@Override
public boolean equals(Object obj) {
    if (obj == this) {
        return true;
    }
    if (!(obj instanceof Media)) {
        return false;
    }
    return ((Media)obj).getTitle() == this.getTitle();
}
```

- Lớp **Track**

```
@Override
public boolean equals(Object obj) {
    if (obj == this) {
        return true;
    }
    if(!(obj instanceof Track)) {
        return false;
    }
    return ((Track)obj).getTitle() == this.getTitle() && ((Track)obj).getLength() == this.getLength();
}
```

- Cập nhật lại lớp **Cart** để kiểm tra phần tử trùng lặp

```
public void addMedia(Media media) {
    if (itemsOrdered.size() >= MAX_NUMBERS_ORDERED) {
        System.out.println("The cart is almost full!");
    } else {
        itemsOrdered.add(media);
        System.out.println(media.getTitle() + "has been added!");
    }
}
```

- Cập nhật lớp **CartTest** để kiểm tra

```
DigitalVideoDisc dvd3 = new DigitalVideoDisc("Aladin",
        "Animation", 18.99f);
cart.addMedia(dvd3);

DigitalVideoDisc dvd4 = new DigitalVideoDisc("Aladin",
        "Animation", 18.99f);
cart.addMedia(dvd4);
```

9. Tính đa hình với phương thức toString()

Branch: feature/aims-project/polymorphism-with-toString-method

- Tùy vào từng loại đối tượng, override phương thức **toString()** thì sẽ hiện ra thông tin khác nhau tùy thuộc vào **toString()** tương ứng.

Lớp **Book**

```java
@Override
public String toString() {
    return "Book: " + this.getTitle() +
           " - Category: " + this.getCategory() +
           " - Cost: " + this.getCost() + "$";
}
```

Lớp **Media**

```java
@Override
public String toString() {
    return "Media: " + this.getTitle() +
           " - Category: " + this.getCategory() +
           " - Cost: " + this.getCost() + "$";
}
```

Lớp **DigitalVideoDisc**

```java
@Override
public String toString()
{
    return "DVD: " + this.getTitle() +
           " - Category: " + this.getCategory() +
           " - Director: " + this.getDirector() +
           " - DVD length: " + this.getLength() +
           " - Cost: " + this.getCost() + "$";
}
```

Lớp **CompactDisc**

```java
@Override
public String toString() {
    return "CD: " + this.getTitle() +
           " - Category: " + this.getCategory() +
           " - Artist: " + this.getArtist() +
           " - Length: " + this.getLength() +
           " - Cost: " + this.getCost() + "$";
}
```

- Kết quả chạy lớp **CartTest**

```java
public class CartTest {
    public static void main(String[] args) {

        Cart cart = new Cart();

        DigitalVideoDisc dvd1 = new DigitalVideoDisc("The Lion King", "Animation", "Roger Allers", 87, 19.95f);
        cart.addMedia(dvd1);

        DigitalVideoDisc dvd2 = new DigitalVideoDisc("Star War", "Science Fiction", "George Lucas", 87, 24.95f);
        cart.addMedia(dvd2);

        DigitalVideoDisc dvd3 = new DigitalVideoDisc("Aladin", "Animation", 18.99f);
        cart.addMedia(dvd3);

        DigitalVideoDisc dvd4 = new DigitalVideoDisc("Aladin", "Animation", 18.99f);
        cart.addMedia(dvd4);

        cart.print();

        cart.searchByID(3);
        cart.searchByTitle("Lion");
    }
}
```

```
The Lion Kinghas been added!
Star Warhas been added!
Aladinhas been added!
Aladin is already in the cart!
*************************CART*************************
Ordered Items:
1.DVD: The Lion King - Category: Animation - Director: Roger Allers - DVD length: 87 - Cost: 19.95$
2.DVD: Star War - Category: Science Fiction - Director: George Lucas - DVD length: 87 - Cost: 24.95$
3.DVD: Aladin - Category: Animation - Director: null - DVD length: 0 - Cost: 18.99$
Total cost: 63.89
*****************************************************
FoundDVD: Aladin - Category: Animation - Director: null - DVD length: 0 - Cost: 18.99$
FoundDVD: The Lion King - Category: Animation - Director: Roger Allers - DVD length: 87 - Cost: 19.95$
```

10. Sắp xếp Media trong Cart

Branch: feature/aims-project/sort-media-in-cart-with-comparator

Sử dụng **Comparator**

- Tạo 2 lớp **Comparator**

MediaComparatorByCostTitle.java

```java
package hust.soict.hedspi.aims.media;

import java.util.Comparator;

public class MediaComparatorByCostTitle implements Comparator<Media>{

    @Override
    public int compare(Media o1, Media o2) {

        int costComparison = Double.compare(o2.getCost(), o1.getCost());
        if (costComparison != 0) {
            return costComparison;
        }

        return o1.getTitle().compareTo(o2.getTitle());
    }
}
```

MediaComparatorByCostTitle.java MediaComparatorByTitleCost.java

```java
package hust.soict.hedspi.aims.media;

import java.util.Comparator;

public class MediaComparatorByTitleCost implements Comparator<Media> {

    @Override
    public int compare(Media o1, Media o2) {

        int titleComparison = o1.getTitle().compareTo(o2.getTitle());
        if (titleComparison != 0) {
            return titleComparison;
        }

        return Double.compare(o2.getCost(), o1.getCost());
    }

}
```

- Thêm các **Comparator** làm thuộc tính của lớp **Media**

```java
public abstract class Media implements Comparable<Media>{

    public static final Comparator<Media> COMPARE_BY_TITLE_COST = new MediaComparatorByTitleCost();
    public static final Comparator<Media> COMPARE_BY_COST_TITLE = new MediaComparatorByCostTitle();
```

- Tạo lớp **MediaCompareTest** để kiểm tra

```java
Iterator<Media> iterator = mediae.iterator();
// Sort by title using comparator
System.out.println();
System.out.println("**********SORT BY TITLE USING COMPARATOR**********");
Collections.sort((List<Media>)mediae, Media.COMPARE_BY_TITLE_COST);
iterator = mediae.iterator();

while (iterator.hasNext()) {
    System.out.println(((Media)iterator.next()).toString());
}
System.out.println("**************************************************");


// Sort by cost using comparator
System.out.println();
System.out.println("**********SORT BY COST USING COMPARATOR************");
Collections.sort((List<Media>)mediae, Media.COMPARE_BY_COST_TITLE);
iterator = mediae.iterator();

while (iterator.hasNext()) {
    System.out.println(((Media)iterator.next()).toString());
}
System.out.println("**************************************************");
```

Phần so sánh **Comparator** và interface **Comparable** sẽ được triển khai ở phần III. Answer Question

11. Tạo Console hoàn chỉnh trong lớp Aims

Branch: topic/aims-project/complete-console-application

- Hàm **main**

```java
public class Aims {

    private static Store store = new Store();
    private static Cart cart = new Cart();

    public static void main(String[] args) {
        // Init add media to the store
        initSetup();
        boolean exit = false;
        //CLI
        while (exit == false)
        {
            showMenu();

            Scanner scanner = new Scanner(System.in);
            int option = scanner.nextInt();
            scanner.nextLine();

            switch (option) {
                case 0:
                    exit = true;
                    System.out.println("Good bye!");
                    break;
                case 1:
                    clearConsole();
                    storeMenu(scanner);
                    break;
                case 2:
                    clearConsole();
                    updateStoreMenu(scanner);
                    break;
                case 3:
                    clearConsole();
                    cartMenu(scanner);
                    break;
                default:
                    clearConsole();
                    System.out.println("Invalid option, please choose again.");
                    break;
            }
        }
```

- Clear Console

```java
// Clear Console
public static void clearConsole() {
    for (int i=0; i<50; i++)
    {
        System.out.println();
    }
}
```

- Khởi tạo Store gồm các Media (CD-Track, DVD, Book)

```java
public static void initSetup() {

    DigitalVideoDisc dvd1 = new DigitalVideoDisc("The Lion King", "Animation", "Roger Allers", 87, 19.95f);
    DigitalVideoDisc dvd2 = new DigitalVideoDisc("Star War", "Science Fiction", "George Lucas", 87, 24.95f);
    DigitalVideoDisc dvd3 = new DigitalVideoDisc("Aladin", "Animation", 18.99f);
    store.addMedia(dvd1);
    store.addMedia(dvd2);
    store.addMedia(dvd3);

    Book book = new Book("The Valley of Fear", "Detective", 20.00f);
    Book book1 = new Book("A Living Remedy: A Memoir", "Biography", 202.00f);
    Book book2 = new Book("On the Origin of Time: Stephen Hawking's Final Theory", "Science", 120.00f);
    store.addMedia(book);
    store.addMedia(book1);
    store.addMedia(book2);

    CompactDisc cd1 = new CompactDisc("Adele - 30", "Music","Adele", 1500.98f);
    Track track1CD1 = new Track("All Night Parking (interlude)", 161);
    Track track2CD1 = new Track("To Be Loved", 403);
    Track track3CD1 = new Track("Woman Like Me", 300);
    cd1.addTrack(track1CD1);
    cd1.addTrack(track2CD1);
    cd1.addTrack(track3CD1);

    CompactDisc cd2 = new CompactDisc("The Gods We Can Touch", "Music","Aurora", 2000.22f);
    Track track1CD2 = new Track("Everything Matters", 180+34);
    Track track2CD2 = new Track("Blood in the Wine", 180+30);
    Track track3CD2 = new Track("Artemis", 60*2+39);
    cd2.addTrack(track1CD2);
    cd2.addTrack(track2CD2);
    cd2.addTrack(track3CD2);

    CompactDisc cd3 = new CompactDisc("Purpose", "Music","Justin Bieber", 1000.98f);
    Track track1CD3 = new Track("The Feeling", 4*60+5);
    Track track2CD3 = new Track("No Sense", 4*60+35);
    cd3.addTrack(track1CD3);
    cd3.addTrack(track2CD3);

    store.addMedia(cd1);
    store.addMedia(cd2);
    store.addMedia(cd3);

    clearConsole();
}
```

- **Menu chính** được gọi trong hàm main()

```java
// Show menu
public static void showMenu() {
    System.out.println("AIMS: ");
    System.out.println("--------------------------------");
    System.out.println("1. View store");
    System.out.println("2. Update store");
    System.out.println("3. See current cart");
    System.out.println("0. Exit");
    System.out.println("--------------------------------");
    System.out.println("Please choose a number: 0-1-2-3");
}
```

- **Menu Store** khi ấn vào View Store

```java
public static void storeMenu(Scanner scanner) {
    boolean back = false;
    while(back == false)
    {
        store.print();

        System.out.println("Options: ");
        System.out.println("--------------------------------");
        System.out.println("1. See a media's details");
        System.out.println("2. Add a media to cart");
        System.out.println("3. Play a media");
        System.out.println("4. See current cart");
        System.out.println("0. Back");
        System.out.println("--------------------------------");
        System.out.println("Please choose a number: 0-1-2-3-4");

        int option = scanner.nextInt();
        scanner.nextLine();

        switch (option) {
            case 0:
                clearConsole();
                back = true;
                break;
            case 1:
                boolean foundDetails = false;
                while (foundDetails == false) {
                    System.out.println("Enter the title of the media (type 0 to stop): ");
                    String title = scanner.nextLine();
                    if (title.equals("0")) {
                        clearConsole();
                        break;
                    }
                    Media media = store.search(title);
                    if (media != null) {
                        clearConsole();
                        System.out.println("Details: ");
                        System.out.println(media);
                        mediaDetailsMenu(scanner, media);
                        foundDetails = true;
                    } else {
                        System.out.println("***MEDIA NOT FOUND***");
                    }
                }
                break;
```

```
                case 2:
                    boolean foundToAdd = false;
                    while (foundToAdd == false) {
                        System.out.println("Enter the title of the media (type 0 to stop): ");
                        String title = scanner.nextLine();
                        if (title.equals("0")) {
                            clearConsole();
                            break;
                        }
                        Media media = store.search(title);
                        if (media != null) {
                            cart.addMedia(media);
                            foundToAdd = true;
                        } else {
                            System.out.println("***MEDIA NOT FOUND***");
                        }
                    }
                    break;
                case 3:
                    boolean foundToPlay = false;
                    while (foundToPlay == false) {
                        System.out.println("Enter the title of the media (type 0 to stop): ");
                        String title = scanner.nextLine();
                        if (title.equals("0")) {
                            clearConsole();
                            break;
                        }
                        Media media = store.search(title);
                        if (media != null) {
                            if (media instanceof Disc || media instanceof CompactDisc) {
                                media.play();
                            } else {
                                System.out.println("This type of media is not supported!");
                            }
                            foundToPlay = true;
                        } else {
                            System.out.println("***MEDIA NOT FOUND***");
                        }
                    }
                    break;
```

```
            case 4:
                clearConsole();
                cartMenu(scanner);
                break;
            default:
                clearConsole();
                System.out.println("Invalid option, please choose again.");
                break;
        }
    }
}
```

- Tùy chọn **See a media's details**

```java
// Media Details Menu
public static void mediaDetailsMenu(Scanner scanner, Media media) {
    boolean back = false;
    while (back == false) {
        System.out.println("Options: ");
        System.out.println("--------------------------------");
        System.out.println("1. Add to cart");
        System.out.println("2. Play");
        System.out.println("0. Back");
        System.out.println("--------------------------------");
        System.out.println("Please choose a number: 0-1-2");

        int option = scanner.nextInt();
        scanner.nextLine();

        switch (option) {
            case 0:
                clearConsole();
                back = true;
                break;
            case 1:
                cart.addMedia(media);
                break;
            case 2:
                if (media instanceof Disc || media instanceof CompactDisc) {
                    media.play();
                } else {
                    System.out.println("This type of media is not supported!");
                }
                break;
            default:
                clearConsole();
                System.out.println("Invalid option, please choose again.");
                break;
        }
    }
}
```

- Từ menu chính chọn **See current cart**

```java
// CartMenu
public static void cartMenu(Scanner scanner) {
    boolean back = false;
    while (back == false) {
        cart.print();
        System.out.println("Options: ");
        System.out.println("--------------------------------");
        System.out.println("1. Filter medias in cart");
        System.out.println("2. Sort medias in cart");
        System.out.println("3. Remove media from cart");
        System.out.println("4. Play a media");
        System.out.println("5. Place order");
        System.out.println("0. Back");
        System.out.println("--------------------------------");
        System.out.println("Please choose a number: 0-1-2-3-4-5");
        int option = scanner.nextInt();
        scanner.nextLine();
        switch (option) {
            case 0:
                clearConsole();
                back = true;
                break;
```

```java
            case 1:
                System.out.println("Filter medias in cart by (1) id or (2) title:");
                int filterOption = scanner.nextInt();
                scanner.nextLine();
                boolean found = false;
                while (found == false) {
                    if (filterOption == 1) {
                        System.out.println("Enter the id to filter (type 0 to stop):");
                        int id = scanner.nextInt();
                        scanner.nextLine();
                        if (id == 0) {
                            clearConsole();
                            break;
                        }
                        cart.searchByID(id);
                        found = true;
                    } else if (filterOption == 2) {
                        System.out.println("Enter the title to filter (type 0 to stop):");
                        String title = scanner.nextLine();
                        if (title.equals("0")) {
                            clearConsole();
                            break;
                        }
                        cart.searchByTitle(title);
                        found = true;
                    } else if (filterOption == 0) {
                        clearConsole();
                        break;
                    } else {
                        System.out.println("Invalid option.");
                    }
                }
                break;
```

```java
            case 2:
                System.out.println("Sort medias in cart by (1) title or (2) cost:");
                int sortOption = scanner.nextInt();
                scanner.nextLine();
                if (sortOption == 1) {
                    cart.sortMediaByTitle();
                } else if (sortOption == 2) {
                    cart.sortMediaByCost();
                } else {
                    System.out.println("Invalid option.");
                }
                break;
            case 3:
                boolean foundToRemove = false;
                while (foundToRemove==false) {
                    System.out.println("Enter the title of the media (type 0 to stop): ");
                    String title = scanner.nextLine();
                    if (title.equals("0")) {
                        clearConsole();
                        break;
                    }
                    Media media = cart.searchToRemove(title);
                    if (media != null) {
                        clearConsole();
                        cart.removeMedia(media);
                        foundToRemove = true;
                    } else {
                        System.out.println("***MEDIA NOT FOUND***");
                    }
                }
                break;
```

```java
            case 4:
                boolean foundToPlay = false;
                while (foundToPlay == false) {
                    System.out.println("Enter the title of the media (type 0 to stop): ");
                    String title = scanner.nextLine();
                    if (title.equals("0")) {
                        clearConsole();
                        break;
                    }
                    Media media = store.search(title);
                    if (media != null) {
                        if (media instanceof Disc || media instanceof CompactDisc) {
                            media.play();
                        } else {
                            System.out.println("This type of media is not supported!");
                        }
                        foundToPlay = true;
                    } else {
                        System.out.println("***MEDIA NOT FOUND***");
                    }
                }
                break;
            case 5:
                clearConsole();
                cart.empty();
                break;
            default:
                clearConsole();
                System.out.println("Invalid option, please choose again.");
                break;
        }
    }
}
```

- Khi người dùng chọn **update store**

```java
// Update Store Menu
public static void updateStoreMenu(Scanner scanner) {
    boolean back = false;
    while (!back) {
        System.out.println("Options: ");
        System.out.println("--------------------------------");
        System.out.println("1. Add a media");
        System.out.println("2. Remove a media");
        System.out.println("0. Back");
        System.out.println("--------------------------------");
        System.out.println("Please choose a number: 0-1-2");
        int option = scanner.nextInt();
        scanner.nextLine();
        switch (option) {
            case 0:
                clearConsole();
                back = true;
                break;
```

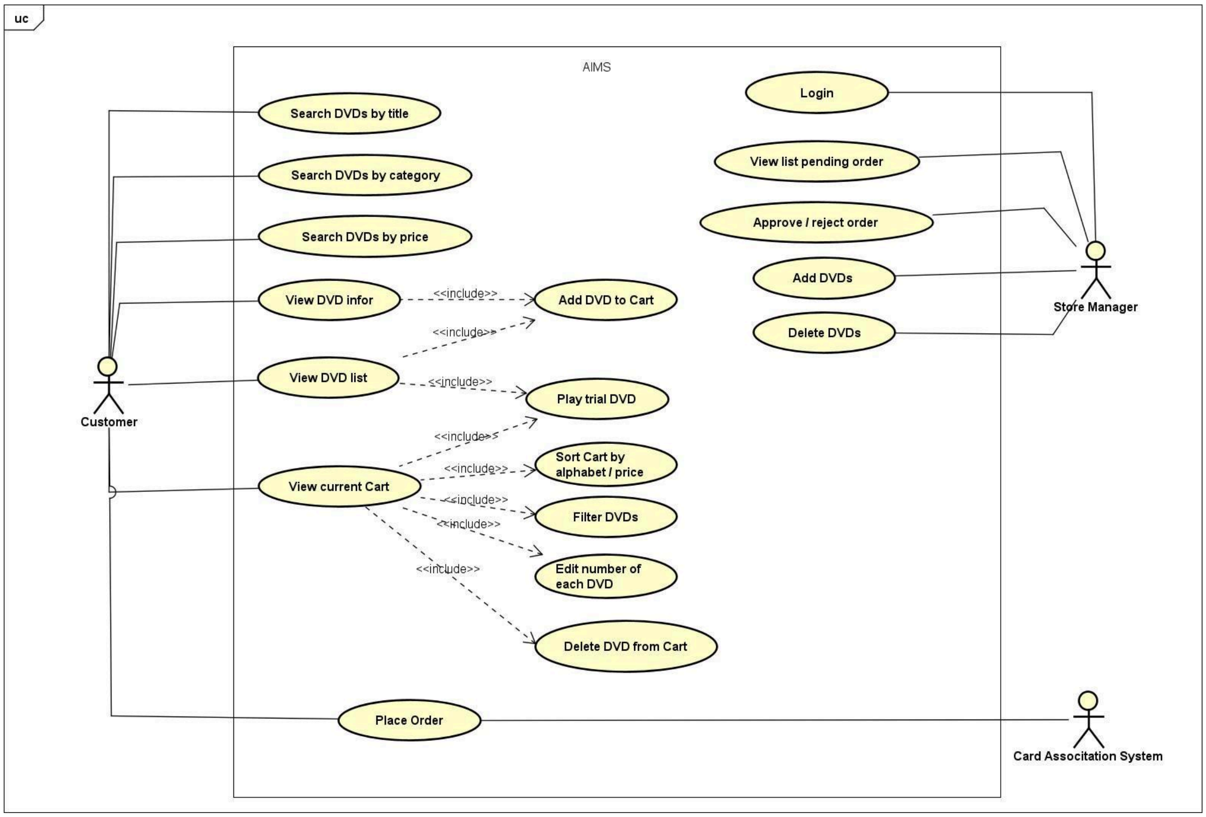
```java
            case 1:
                System.out.println("Enter the category of the media (1) Book, (2) CD, (3) DVD or (0) exit:");
                int categoryChoice = scanner.nextInt();
                scanner.nextLine();

                if (categoryChoice == 1) {
                    System.out.println("Enter book title: ");
                    String bookTitle = scanner.nextLine();
                    System.out.println("Enter book category: ");
                    String bookCategory = scanner.nextLine();
                    System.out.println("Enter book cost: ");
                    Float bookCost = scanner.nextFloat();
                    scanner.nextLine();

                    Book newBook = new Book(bookTitle, bookCategory, bookCost);
                    store.addMedia(newBook);
                } else if (categoryChoice == 2) {
                    System.out.println("Enter CD title: ");
                    String cdTitle = scanner.nextLine();
                    System.out.println("Enter CD category: ");
                    String cdCategory = scanner.nextLine();
                    System.out.println("Enter CD artist: ");
                    String cdArtist = scanner.nextLine();
                    System.out.println("Enter CD cost: ");
                    Float cdCost = scanner.nextFloat();
                    scanner.nextLine();

                    CompactDisc newCD = new CompactDisc(cdTitle, cdCategory, cdArtist, cdCost);


                    System.out.println("Do you want to add tracks to your CD? (1) Yes (0) No:");
                    int addTrack = scanner.nextInt();
                    scanner.nextLine();
```

```java
                    if (addTrack == 1) {
                        System.out.println("How many tracks in your CD?");
                        int numTrack = scanner.nextInt();
                        scanner.nextLine();
                        for (int i = 0; i < numTrack; i++) {
                            System.out.println("Your " + (i+1) + " track: ");
                            System.out.println("Enter track title: ");
                            String trackTitle = scanner.nextLine();
                            System.out.println("Enter track length: ");
                            int trackLength = scanner.nextInt();
                            scanner.nextLine();

                            Track newTrack = new Track(trackTitle, trackLength);
                            newCD.addTrack(newTrack);
                        }
                        store.addMedia(newCD);
                    } else if (addTrack == 0) {
                        store.addMedia(newCD);
                    }
                } else if (categoryChoice == 3) {
                    System.out.println("Enter DVD title: ");
                    String dvdTitle = scanner.nextLine();
                    System.out.println("Enter DVD category: ");
                    String dvdCategory = scanner.nextLine();
                    System.out.println("Enter book cost: ");
                    Float dvdCost = scanner.nextFloat();
                    scanner.nextLine();

                    DigitalVideoDisc newDVD = new DigitalVideoDisc(dvdTitle, dvdCategory, dvdCost);
                    store.addMedia(newDVD);
                } else if (categoryChoice == 0) {
                    clearConsole();
                    break;
                } else {
                    System.out.println("Invalid option.");
                }
                break;
```

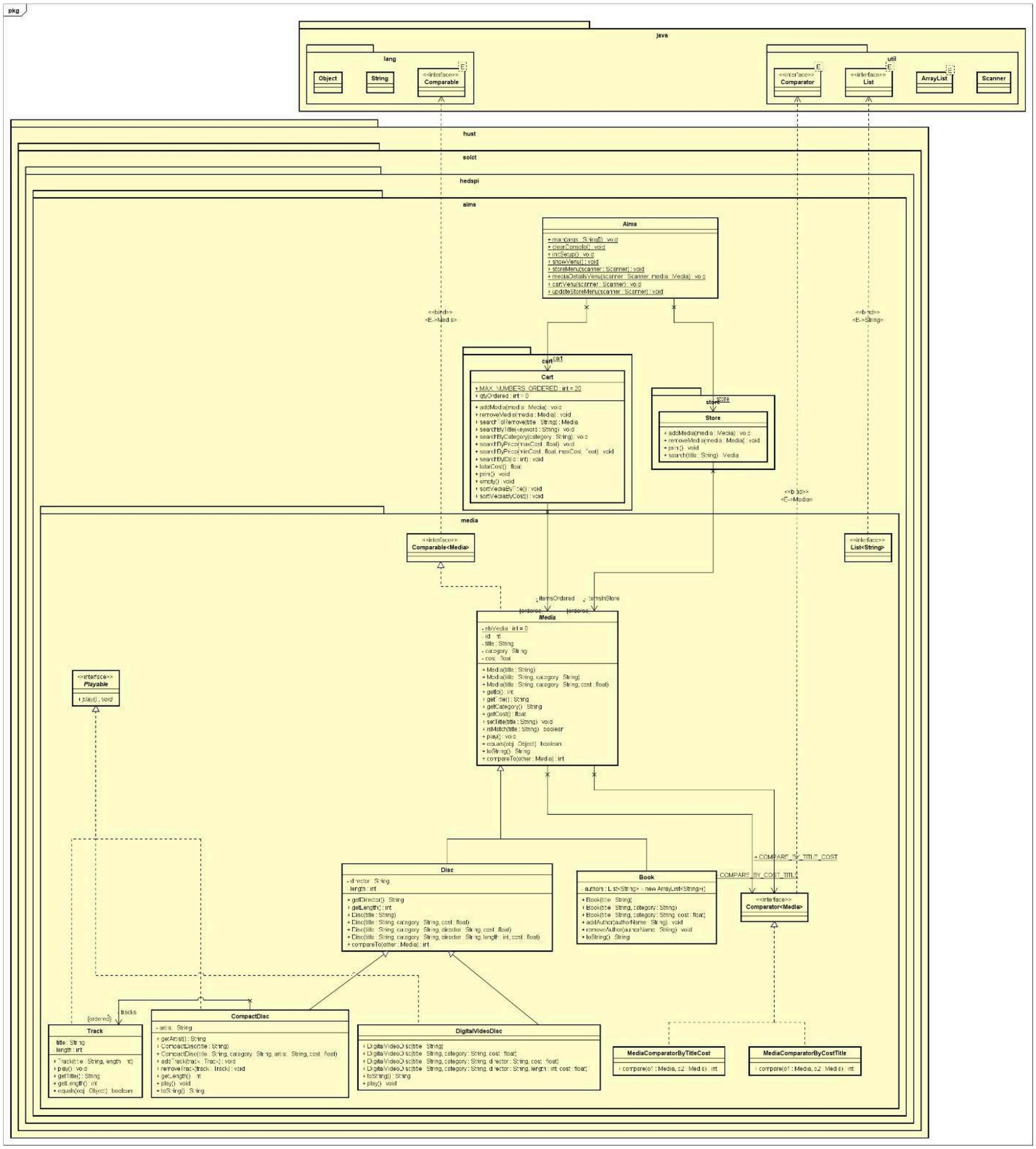
```java
                case 2:
                    boolean foundToRemove = false;
                    while (!foundToRemove) {
                        System.out.println("Enter the title of the media (type 0 to stop): ");
                        String titleForRemove = scanner.nextLine();
                        if (titleForRemove.equals("0")) {
                            clearConsole();
                            break;
                        }
                        Media media = store.search(titleForRemove);
                        if (media != null) {
                            clearConsole();
                            store.removeMedia(media);
                            foundToRemove = true;
                        } else {
                            System.out.println("***MEDIA NOT FOUND***");
                        }
                    }
                    break;
                default:
                    clearConsole();
                    System.out.println("Invalid option, please choose again.");
                    break;
            }
        }
    }
}
```

II. UML Diagram

1. Use-Case Diagram



2. Class Diagram (Final Lab04)



III. Answer Question

Question: Alternatively, to compare items in the cart, instead of using **Comparator**, we can use the **Comparable** interface and override the **compareTo()** method. You can refer to the Java docs to see the information of this interface.

Suppose we are taking this **Comparable** interface approach.

1. *What class should implement the Comparable interface?*

 Media class should implement the **Comparable** interface

```
public abstract class Media implements Comparable<Media>{
```

2. *In those classes, how should you implement the compareTo()method be to reflect the ordering that we want?*

The answer can be found in **src/hust/socit/hedspi/aims/Media.java** file.

```
@Override
public int compareTo(Media other) {
    int titleComparison = this.getTitle().compareTo(other.getTitle());
    if (titleComparison != 0) {
        return titleComparison;
    } else {
        return Double.compare(this.getCost(), other.getCost());
    }
}
```

3. *Can we have two ordering rules of the item (by title then cost and by cost then title) if we use this Comparable interface approach?*

No we cannot. The **Comparable** interface assumes that there is only one natural ordering for the objects being compared.

4. *Suppose the DVDs has a different ordering rule from the other media types, that is by title, then decreasing length, then cost. How would you modify your code to allow this?*

We can override the **compareTo()** method in **Disc** class to reflect the new ordering rule.

The modified code can be found in **src/hust/soict/hedspi/aims/Disc.java** file

```java
    @Override
    public int compareTo(Media other) {
        if (other instanceof Disc) {
            Disc otherDVD = (Disc) other;
            int titleComparison = this.getTitle().compareTo(otherDVD.getTitle());
            if (titleComparison != 0) {
                // Compare by title
                return titleComparison;
            } else {
                // Compare by decreasing length
                int lengthComparison = Integer.compare(otherDVD.getLength(), this.getLength());
                if (lengthComparison != 0) {
                    return lengthComparison;
                } else {
                    // Compare by cost
                    return Double.compare(this.getCost(), otherDVD.getCost());
                }
            }
        } else {
            // If the media object is not a Disc, use the default method of the Media class
            return super.compareTo(other);
        }
    }
```

MediaCompareTest

```java
    // Sort using override compareto method
    System.out.println();
    System.out.println("*******SORT USING OVERRIDE compareTo method*******");
    Collections.sort(mediae);
    Iterator<Media> iterator = mediae.iterator();
    while (iterator.hasNext()) {
        System.out.println(((Media)iterator.next()).toString());
    }

    System.out.println("***************************************************");
```